Closed Native Verification Systems (CNVS): Complete Formal Theory and Emergent Security Theorem

GENERAL INDEX

Abstract

This paper introduces Closed Native Verification Systems (CNVS), a formal class of verification systems in which state membership is verification-dependent rather than primitive. The theory combines a typed structural universe, a type-preserving recursive decomposition, a random assignment of non-uniform fragments, limited verifier knowledge, and a global invariant check, first applying data convergence in a local verification environment. A conditional probabilistic theorem limits unauthorized reconstruction with a binomial tail and produces exponential suppression when the reconstruction threshold exceeds the expected number of useful compromised fragments. An asymptotic scaling theorem shows that security can increase as the system expands in the presence of explicit assumptions.
The technical difficulty of reconstruction outside the system arises from the strictly local knowledge available to each verifier.

1. Primitive Domains and Typed Universe

Let $B = \{0,1\}$ and B^* be the set of all finite binary strings. Define the primitive domains:
$Id \subseteq B^*$ (identifiers),
$D \subseteq B^*$ (declared data),
$At \subseteq B^*$ (attribute classes),
$Val \subseteq B^*$ (externally reported values),
P (finite set of logical properties),
Tau (finite set of structural types).
The structural universe is:
$U = Id \times D \times At \times Val \times Pow(P)$.
Each element $s \in U$ has the form $s = (id, d, a, v, p)$.

2. State Space

A system state at time t is:
$S(t) = (E(t), R(t), C, V\_L)$,
where $E(t) \subseteq U$ is the set of admitted elements, $R(t) \subseteq E(t) \times E(t)$ is a directed relation set, $C = \{c1, \ldots, cm\}$ is a finite family of invariant predicates, and V_L is the local verification function.

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3. Local Verification

The local verification function is:

$V\_L : Id \times D \times At \times Val \to \{0,1\} \times Pow(P)$.

For $x = (id, d, a, v)$*, let:*

$V\_L(x) = (b, p)$,

where

$b = Form(id,d,a,v) \wedge Conv\_a(d,v)$.

Local admissibility is:

$Adm\_L(s,t) = 1 \Leftrightarrow \pi 1(V\_L(s)) = 1$.

4. Recursive Decomposition and Reconstruction

The decomposition operator is:

$D\_t : U \to Pow\_fin(U)$.

For every admissible element s:

$D\_t(s) = \{F1, \ldots, Fk\}, 1 \leq k < \infty$.

Type preservation:

$\forall s' \in D\_t(s), type(s') = type(s)$.

Reconstruction:

$Rec\_t(D\_t(s)) = s$

up to canonical encoding.

5. Non-Uniform Fragmentation

There exist fragments Fa and Fb such that:

$|I(Fa)| \neq |I(Fb)|$.

Hence decomposition is intentionally non-uniform. Some fragments contain more information than others.

6. Knowledge Restriction

For each external verifier v_j assigned to attribute a_j:

$I_{min}(a_j) <= K_V(j,t) << I_N(i,t) << I_G(t)$

where $I_{min}(a_j)$ is the minimum information required to preserve the semantic meaning of the assigned verification task.

Knowledge cannot converge to zero absolutely, since complete loss of information would eliminate the verifier's ability to perform the assigned task.

7. Information Density

For a fragment F_j:

$\rho_N(j,i,t) = |I_F(j,t)| / |I_N(i,t)|$,

$\rho_G(j,t) = |I_F(j,t)| / |I_G(t)|$,

$\rho_C(j,i,t) = \rho_N(j,i,t)^\alpha \rho_G(j,t)^\beta$,

where $\alpha,\beta \in [0,1]$ and $\alpha + \beta = 1$.

8. Random Assignment

Terminal fragments are assigned to external verifiers by:

$A_t : T(t) \times \Xi \rightarrow V_{ext}(t)$,

where Ξ is a probability space and $T(t)$ is the set of terminal fragments.

9. Axiom I: Verification-Dependent Existence

For every element s and time t:

$s \in E(t) \Rightarrow Adm_L(s,t) = 1$.

Existence inside the system is a derived consequence of successful verification.

10. Axiom II: Randomized Non-Uniform Fragmentation

Every admissible element is recursively decomposed into non-uniform fragments that are assigned randomly:
$D_t(s) = \{F1, \ldots, Fk\}$,
$\exists Fa,Fb : |I(Fa)| \neq |I(Fb)|$,
$A_t : T(t) \times \Xi \rightarrow V_{ext}(t)$,
$Rec_t(D_t(s)) = s$.

11. Axiom III: Non-Reducibility of Global Validity

Global validity is not reducible to local validity:
$V_G(S) \neq \wedge_{\{s\in E\}} V_L(s)$.
Instead:
$V_G(S) = LocalOK(E) \wedge Cons_R(R,E) \wedge Inv_C(S)$.

12. Fundamental Lemmas

Lemma 1. Local admissibility is necessary for state membership.
Lemma 2. Local validity is insufficient for global validity.
Lemma 3. Type is preserved across decomposition depth.
Lemma 4. Reconstruction conserves payload information.
Lemma 5. Metadata overhead permits encoded expansion.
Lemma 6. Average fragment density decreases under controlled subdivision.

13. Principle of Knowledge Restriction

For every external verifier:
$I_{min}(a_j) <= K_V(j,t) << I_G(t)$.
If K_V remains bounded while $|I_G|$ grows, then:
$|K_V(j,t)| / |I_G(t)| -> 0$.

Thus each verifier retains the minimum semantic information required to perform the assigned task while controlling an asymptotically vanishing fraction of the total global information.

14. Probabilistic Security Model

Let:
k = *number of terminal fragments,*
r = *reconstruction threshold,*
C = *colluding verifier count,*
q = *total verifier count,*
$p = C/q$,
λ_j = *probability that a compromised fragment is useful,*
$p_{eff} = p\,\lambda_{max}$.
Define $X_j = 1$ *if fragment* j *is compromised and useful, and*
$X = \Sigma_j X_j$.
Unauthorized reconstruction occurs when:
$Rec^* = \{X \geq r\}$.

15. Conditional Security Theorem

If X *is stochastically dominated by* $Bin(k, p_{eff})$, *then:*
$Pr(Rec^*) \leq \Sigma_{i=r}^{k} \binom{k}{i} p_{eff}^i (1-p_{eff})^{(k-i)}$.

Proof.
By definition, unauthorized reconstruction requires at least r *useful compromised fragments.*
Under stochastic domination, $X \leq_{st} Y$ *where* $Y \sim Bin(k,p_{eff})$. *Therefore:*
$Pr(X \geq r) \leq Pr(Y \geq r)$,
which equals the stated binomial upper tail. QED.

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16. Exponential Chernoff Bound

Let $\mu = k\,p\_eff$. If $r > \mu$, then:
$Pr(Rec^*) \le \exp(-(r-\mu)^2 / (2\mu + (r-\mu)))$.

Proof.
Apply the multiplicative Chernoff bound to $Y \sim Bin(k,p\_eff)$. QED.

17. Emergent Security Scaling Theorem

Let n denote system scale. Assume:
$k(n) \to \infty$,
$p\_eff(n)$ remains bounded or decreases,
$r(n) - k(n)p\_eff(n) \to +\infty$.
Then:
$Pr(Rec^*\_n) \to 0$.

Proof.
Substitute the scaling assumptions into the Chernoff exponent. The exponent diverges to $+\infty$, hence the probability tends to zero. QED.

18. Interpretation

Under the stated assumptions, system growth increases fragmentation and epistemic restriction. As a result, unauthorized reconstruction becomes exponentially less probable. Security is therefore an emergent property of scale.

18a. Corollary: Global Veto Property

If at least one essential contribution to a global invariant remains unknown or uncontrolled by colluding verifiers, reconstruction of a globally valid state is not guaranteed.

Any incorrect estimation of that contribution may cause one or more global invariants to fail, yielding:

$V_G(S) = 0$
and rejection of the candidate state.

18b. Extreme Collusion Principle

Even if colluding agents control 99.9% of all nodes, a single uncontrolled essential fragment may prevent coherent reconstruction.
Because fragment sizes and semantic weights are non-uniform, the unknown contribution cannot be inferred with certainty.
If the guessed contribution is incorrect, one or more global invariants fail and the entire candidate state is rejected.

18c. Gold Allocation Example

Suppose 100 orders should sum to 100 grams of gold.
If colluding agents know 99 orders totaling 99 grams, the remaining unknown order may represent any admissible quantity.
An incorrect estimate may produce a total inconsistent with the invariant, for example 102 grams instead of 100 grams.
Therefore:
$V_G(S) = 0$.
As the value of $|I_G|$ increases, the difficulty of reconstructing the missing information while preserving global invariant consistency increases correspondingly.

19. Scientific Positioning

CNVS combines concepts from information theory, probability theory, cryptography, formal methods, graph theory, and epistemic logic into a unified mathematical framework.

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20. Open Problems

1. Formal leakage models.
2. Dynamic collusion models.
3. Cryptographic reductions.
4. Information-theoretic lower bounds.
5. Empirical validation.

Appendix A. Minimal Formal Definition

A structure
M = (U, E, R, C, D, A, V_L, V_G, Rec, K)
is a Closed Native Verification System if and only if:
1. U is a typed finite-binary universe.
2. Membership requires local admissibility.
3. Global validity requires relation consistency and invariants.
4. Decomposition is finite, recursive, and reconstructible.
5. Verifier knowledge is strictly restricted.
6. Terminal assignments are randomized.
7. Invalid elements and states are rejected.

END DOCUMENT

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